mhs-poisson-ltau: magnetohydrostatic correction on an optical-depth grid

The physics of `run.mode = "mhs-poisson-ltau"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

An inversion gives the temperature T and the magnetic field $\mathbf{B}$ on surfaces of constant optical depth τ_{500}. The mode `column-hse-ltau` turns this into pressures, densities and heights by treating every pixel column as an isolated hydrostatic column. Here we go one step further, as `mhs-poisson-z` does on a height grid: we also require the columns to be in **horizontal force balance with each other** and include the **Lorentz force** of the given field. T and $\mathbf{B}$ on the τ surfaces stay fixed. What is found is the pressure on every τ surface and, at the same time, the geometric height of every τ surface in every column. On a τ grid these two cannot be separated: the geometry is part of the answer.

2. The quantities

Symbol	Meaning	Unit
τ	optical depth at 500 nm; the levels are equally spaced in $\log_{10}\tau$	–
T, $\mathbf{B}$	temperature and field on the τ levels, given, fixed	K, T
P, u	gas pressure on the τ levels; $u = \ln P$	Pa, –
ρ, χ	density and 500 nm extinction, both functions of (T, P) from the equation of state	$\mathrm{kg\,m^{-3}}$, $\mathrm{m^{-1}}$
$z(x, y, \tau)$	height of a τ level at the horizontal position (x, y)	m
τ_c, c	the cut level (default $\log_{10}\tau_c = -4$) and its height in each column	–, m
$\mathbf{J}$, $\mathbf{L}$	current $\nabla\times\mathbf{B}/\mu_0$ and Lorentz force $\mathbf{J}\times\mathbf{B}$, in physical space	$\mathrm{A\,m^{-2}}$, $\mathrm{N\,m^{-3}}$
g, μ_0	gravity 274; vacuum permeability $4\pi\times10^{-7}$	$\mathrm{m\,s^{-2}}$, $\mathrm{T\,m\,A^{-1}}$

3. The force balance

A static gas has zero net force per unit volume:

$$\mathbf{F} = -\nabla P + \mathbf{J}\times\mathbf{B} - \rho g\,\hat{\mathbf{z}} = 0, \qquad \mathbf{J}\times\mathbf{B} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0}.$$

The Lorentz force is a magnetic pressure plus a tension along curved field lines. As explained for `mhs-poisson-z`, with the field fixed this balance can in general not be satisfied exactly (the tension of a given field does not hold up the buoyancy of a given temperature structure), so we look for the pressure that balances the forces *as well as possible*. The new difficulty here is that all derivatives in $\mathbf{F}$ are in physical space, at fixed height, whereas our data live on τ surfaces.

4. Heights follow from the pressure

Going down by dz adds the optical depth $d\tau = -\chi\,dz$, so in each column

$$\frac{dz}{d\ln\tau} = -\frac{\tau}{\chi(T,P)}, \qquad z_k = c + \sum_{m=k_c}^{k-1}\Delta z_m, \quad \Delta z_m = -\int_{\ln\tau_m}^{\ln\tau_{m+1}}\frac{\tau}{\chi}\,d\ln\tau.$$

The height of a level is the height c of the cut level plus the thicknesses of the layers in between, and each thickness is set by the opacity, which depends on the pressure. Two consequences:

- *A change of pressure at one level moves every level below it*: higher pressure, more opacity, thinner layer, and everything below shifts up.
- The unknowns are the pressures on the τ levels and one height c per column; all other heights follow.

Above the cut ($\log\tau < -4$) the column is kept hydrostatic in τ, exactly as in `column-hse-ltau`, and moves up and down rigidly with c.

5. Vertical balance along τ

The vertical component of $\mathbf{F} = 0$ is $dP/dz = -(\rho g - L_z)$: the pressure difference across a slab carries its weight minus whatever the vertical Lorentz force carries. Divide by $dz = -d\tau/\chi$ and multiply by τ/P:

$$\boxed{\frac{d\ln P}{d\ln \tau} = \frac{(\rho g - L_z)\,\tau}{\chi\, P}}$$

Without the field this is the hydrostatic equation of `column-hse-ltau`. An upward Lorentz force ($L_z > 0$) carries part of the weight, so the pressure needs to rise less steeply with optical depth.

6. Horizontal balance on tilted surfaces

A τ surface is generally not flat: its height $z(x, y, \tau)$ varies across the box. Consider the pressure along such a surface, $\tilde{P}(x, \tau) = P(x, z(x, \tau))$. By the chain rule, moving along x on the surface changes it by

$$\left.\frac{\partial P}{\partial x}\right|_{\tau} = \left.\frac{\partial P}{\partial x}\right|_{z} + \frac{\partial P}{\partial z}\left.\frac{\partial z}{\partial x}\right|_{\tau}.$$

The physical force needs $\partial P/\partial x$ at fixed height, so we solve for it and use the vertical balance $\partial P/\partial z = -(\rho g - L_z)$:

$$\boxed{F_x = -\left.\frac{\partial P}{\partial x}\right|_{\tau} - (\rho g - L_z)\left.\frac{\partial z}{\partial x}\right|_{\tau} + L_x}$$

and the same for y. The middle term has a simple meaning. If a τ surface slopes, comparing pressures along it means comparing points at different heights, and the weight of the gas between them must be subtracted. On a tilted surface part of the weight acts sideways. Horizontal balance therefore ties together the pressures *and* the heights of neighbouring columns: a pressure difference along a τ surface can be balanced by a tilt of the surface.

7. The Lorentz force on a τ grid

$\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$ also needs derivatives in physical space. With the field given on the τ levels, the same chain rule gives, for any field component f,

$$\frac{\partial f}{\partial z} = \frac{df/d\ln\tau}{dz/d\ln\tau}, \qquad \left.\frac{\partial f}{\partial x}\right|_{z} = \left.\frac{\partial f}{\partial x}\right|_{\tau} - \left.\frac{\partial z}{\partial x}\right|_{\tau}\frac{\partial f}{\partial z}.$$

So even though the field values on the τ levels are fixed, the currents and the Lorentz force depend on where the levels lie: if two levels move apart, the vertical gradients of the field between them become smaller, and so do the currents they imply. The solver accounts for this dependence.

8. The quantity minimised

As on the height grid, the forces are divided by the pressure (a relative imbalance, so that lowering the pressure is not rewarded) and weighted so that each layer counts in units of the weight of its own gas:

$$R_x = \frac{F_x}{P}, \quad R_y = \frac{F_y}{P}, \quad R_z = \Delta\ln P - \int \frac{(\rho g - L_z)\tau}{\chi P}\, d\ln\tau, \qquad \Phi = \sum_{\text{levels}} w\,\bar{H}^2\left(R_x^2 + R_y^2\right) + \sum_{\text{intervals}} \frac{\bar{H}^2}{\Delta\bar{z}} R_z^2.$$

$\bar{H}$ is the typical scale height $P/(\rho g)$ of a level and w, $\Delta\bar{z}$ its typical thickness, so that $\bar{H}R \approx F/(\rho g)$ and Φ approximates a volume integral of the squared imbalance in units of the weight. Φ is minimised over the pressures below the cut and the column heights c (least squares: three equations per point, one pressure unknown per point and one height per column, no exact solution).

9. What decides the heights: the Wilson depression

Consider a column with a strong, nearly vertical field among weak-field neighbours. Its magnetic pressure $B^2/2\mu_0$ pushes outward on the neighbours, so at the same height it can hold its own against them with less gas pressure. Less gas pressure means lower density and lower opacity, so light escapes from deeper: the τ surfaces of this column lie lower. Horizontal balance favours exactly this through the tilt term of Sect. 6. The depression of strong-field regions, seen in sunspots near the limb as the **Wilson depression**, is therefore expected to come out of the force balance rather than being put in by hand, although for a given column the result also depends on its temperature structure and on how well the compromise balances the forces there. Conversely, hot granules have a high opacity, and their τ surfaces tend to lie higher.

10. What is determined, and what is not

- **The global height.** All columns can be moved up or down together without changing any force. This offset is fixed at the end by setting the average height of $\tau = 1$ to zero, a convention, not a measurement.
- **Above the cut** ($\log \tau < -4$ by default) the levels are not solved: their pressures and their distances to the cut level stay those of the hydrostatic columns in τ, and the whole upper part moves rigidly with the cut level.
- **Below the cut** the pressures and the heights are determined only as well as the least-squares compromise determines them. A few strong-field columns are weakly constrained: their heights can change noticeably while Φ barely changes.
- **The magnetic field is not made consistent with the new geometry.** The field values on the τ levels are kept, while the levels move in space. Nothing enforces $\nabla \cdot \mathbf{B} = 0$ for the field at its new positions; the divergence-free construction of `full-mhs-z` (a field built as the curl of a potential on a fixed grid) does not carry over to a grid whose geometry is itself an unknown. The Lorentz force is simply computed from the field as it lies.

11. How the minimum is found

Φ depends non-linearly on the unknowns (through the equation of state, the opacity in the heights and the field derivatives). Each step replaces the residuals by their linear approximation in the change of the unknowns (Gauss–Newton, the least-squares form of Newton's method) and solves the resulting linear least-squares problem. The linear problem contains the cumulative sum of Sect. 4 (a pressure change shifts all heights below it), a horizontal coupling between neighbouring columns (Sect. 6), and the vertical equation of Sect. 5. It is solved iteratively, guided by a fast approximate solution in horizontal Fourier modes, as on the height grid.

A pure Gauss–Newton step can be too ambitious where the problem is strongly non-linear. Two safeguards keep it reliable. First, the step is scaled down as a whole if it would move any column by more than 300 km or change any pressure by more than a factor e^2; scaling the whole step keeps its direction, which still lowers Φ. Second, a step that does not lower Φ is retried with **damping** (Levenberg–Marquardt): a penalty on the size of the change, which turns the step gradually from the Gauss–Newton direction toward the direction of steepest descent. The iteration stops when Φ has stopped decreasing over several consecutive steps.

12. Summary

On a τ grid the force balance determines both the pressure on each τ surface and where the surfaces lie in space. Vertically the gas is hydrostatic along τ with the vertical Lorentz force carrying part of the weight; horizontally the columns balance each other across tilted surfaces; the heights follow from the opacity. As on the height grid, a fixed field cannot hold up every density contrast, so the balance is the best possible compromise, and above the cut the columns are only hydrostatic.